

Question ID: ebed7dc6

Question

An auditorium has seats for **1,800** people. Tickets to attend a show at the auditorium currently cost **\$4.00**. For each **\$1.00** increase to the ticket price, **100** fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

Answer

- A. **4**
- B. **7**
- C. **14**
- D. **18**